

Large-Scale Video Analysis and Synthesis using Compositions of Spatiotemporal Labels

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Fig. 1. We compose low-level video labels (faces, human poses, aligned transcripts, etc.) to form high-level spatiotemporal labels that are useful for analysis and synthesis tasks. Left: Understanding representation in the media: screen time of female hosts on cable TV news shows increased steadily since 2010 before leveling off around 2015. Center: Understanding time given to political candidates during on-air interviews. Right: Curating shots of the stars from the Harry Potter series for use in a supercut.

Advances in video processing techniques now make it feasible to automatically annotate video collections with basic information about their audiovisual contents. However, basic annotations such as faces, poses, or time-aligned transcripts, when considered in isolation, offer only a low-level description of a video's contents. In this paper we demonstrate that compositions of these basic annotations are a powerful means to create increasingly higher level labels that are useful for a range of video analysis and synthesis tasks. To facilitate composition of multi-modal data, we represent all annotations as spatiotemporal labels that are associated with continuous volumes of spacetime in a video. We then define high-level labels in terms of spatial and temporal relationships among existing labels. We use these high-level labels to conduct analyses of two large-scale video collections: a database of nearly a decade of 24-7 news coverage from CNN, MSNBC, and Fox News and a collection of films from the past century. These explorations ask questions about the presentation of these sources, such as gender balance

in the media, and the evolution of film editing style. We also leverage label composition for the video synthesis task of curating source clips for video supercuts.

CCS Concepts: • **Computing methodologies** → **Graphics systems and interfaces**.

Additional Key Words and Phrases: video processing, video data mining

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1 INTRODUCTION

Video is our most prolific medium for communicating information and telling stories. From YouTube content to 24/7 cable TV news to feature-length films, we are producing and consuming more video than ever before, so much that streaming video now constitutes almost 60% of worldwide downstream internet bandwidth [Sandvine 2018]. But, because of the sheer volume of media, the public has little understanding of how these stories are told, who is telling them, and how different subjects are presented visually. Yet, these decisions about visual presentation influence how billions receive information and shape public opinion and culture.

For example, in the context of cable TV news, we might ask *How much screen time is afforded to men versus women? Which hosts*

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dominate the screen-time of their respective shows? In the context of films we might similarly ask, *What fraction of film screen time is filled by conversation scenes? What are common shot editing patterns for these scenes?* Analyzing the audiovisual contents of large TV and film video collections stands to provide insight into these editorial decisions. Video analysis is also valuable to those seeking to create new video content. For example, we might find clips of *Barack Obama saying the words “happy birthday”* as a starting point when making the next viral supercut. A film scholar or aspiring actor might benefit from observing the performance of *all reaction shots in Apollo 13 featuring Tom Hanks*.

Today, cloud-scale computation and modern computer vision techniques make it feasible to automatically annotate video with basic information about their audiovisual contents (e.g., faces, genders, poses, transcripts). However, these annotations offer only a low-level description of a video’s contents.

In this paper, we demonstrate that programmatic composition of these basic annotations, which we model as continuous volumes of spacetime within a video, is a powerful way to describe increasingly high-level labels that are useful for a range of video analysis and synthesis tasks. For example, we demonstrate how specific spatial arrangements of face bounding boxes can be used to partition films into shots, how patterns of face appearances over time signal the presence of conversations and interviews, and how per-frame histograms and transcript data can be combined to automatically label TV commercials. Specifically, we make the following contributions:

- We represent multi-modal video annotations using a single common representation, a spatiotemporal label, and demonstrate how high-level labels can be constructed in terms of spatial and temporal relationships of existing labels.
- We use spatiotemporal labels to conduct analyses of two large video collections: nearly a decade of 24-7 news coverage from CNN, MSNBC, and Fox News and a collection of films from the past century. These explorations investigate the presentation of material in these collections, such as gender balance in the media, and the evolution of film editing style.
- We demonstrate use of spatiotemporal labels to select patterns of interest from videos, and use this capability to rapidly curate source clips for synthesizing new video supercuts.

1.1 Video Datasets

Cable TV news. The TV News dataset consists of nearly 24-7 broadcasts from the “big three” U.S. cable news networks (CNN, FOX, and MSNBC) recorded between January 2010 to August 2018 by the Internet Archive [Int 2018]. We process all videos recorded during this period, eliding only videos unusable due to encoding corruption. Videos are H.264 encoded at 480p resolution and also contain original audio, closed caption transcripts, and metadata about the recorded channel, current show name, and time of broadcast. In total, the 83 TB video dataset contains 199,202 hours of video, and spans 200+ unique shows, making it one of the largest video archives ever explored in academic literature.

Feature films. The FILM dataset is a collection of 589 feature-length films spanning release dates 1915 to 2016, and covering a broad range of genres. The films include older, low resolution, movies (e.g.

The Italian (1915), 480×360, 74 minutes) to high-resolution modern blockbusters (e.g. *Interstellar* (2014), 1920×1080, 2 hours 48 minutes). The dataset also includes captions for 97% of the videos retrieved from opensubtitles.co and other sites. Metadata about characters (including leading characters), director, cinematographer, producers, editors, and writers was retrieved from IMDb and Wikipedia. In total, the 2.3 TB dataset contains 1,135 hours of video (99 million frames) from films made between 1915 and 2016. While the dataset spans over a century of film, it is heavily biased towards the years 2011-2016 (59 of the films were released in 2015, see Section 1.2 of the supplemental material for the distribution in time).

2 RELATED WORK

Visual data mining at scale. Mining large image collections is an increasingly popular approach to generating unique insights into the visual world. Examples include characterizing local trends in fashion [Matzen et al. 2017] and evolution of style from yearbook pictures [Ginosar et al. 2017], predicting urban housing prices and crime [Arietta et al. 2014], identifying distinctive visual attributes of cities [Doersch et al. 2012], and using average images as an interface to reveal common structure in internet photographs [Zhu et al. 2014]. Most prior visual data mining efforts have analyzed unstructured image collections. However, in this project, our datasets are multi-modal (video, audio, and text), and are of vastly greater scale.

Representation in media. Components of our TV news and film analyses take inspiration from many efforts to understand representation in modern media [Savillo 2016; Women’s Media Center 2017]. These efforts seek to improve diversity by bringing transparency to the race, gender, and political persuasion of individuals that appear in print or on screen. Most current efforts rely on manual coding of video (human labeling of people, topics, etc.), which limits the amount of media that can be observed as well as the variety of labels collected. Recently, face gender classification and audio analysis was used to estimate female screen time and speaking time in 200 Hollywood films from 2014-2015 [Geena Davis Institute 2016]. Our tools enable a broader set of analysis tasks over a larger video dataset distributed over a century. Much like how Google’s N-gram viewer enabled new studies of culture through analysis of centuries of text [Michel et al. 2011], we hope to augment existing efforts and facilitate new forms of data-driven media analysis at scale.

Film cinematography studies. Cinematographers manipulate elements of a film’s visual style, such as character framing and cut timing, to convey narrative structure, actions, and emotion in scenes. Scholars have examined these choices to characterize trends in cinema [Bordwell 2006; Brunick et al. 2013; Cutting 2016; Cutting et al. 2011]. For example, Cutting et al. [2011] manually annotate low-level video features (shot boundaries, shot framings, actor head positions) to quantify trends such as the increase in the number of close up shots and in camera motion over time. We replicate many of Cutting’s findings using automated techniques, enabling analysis of a significantly larger set of films.

Film browsing and search. Pavel et al. [2015] developed an interface for searching and browsing film collections using time-aligned captions, scripts [Ronfard and Thuong 2003], and plot summaries.

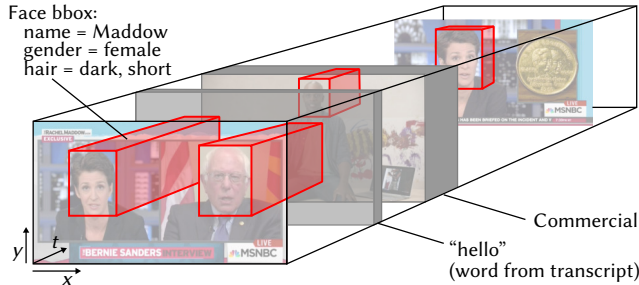


Fig. 2. We represent all labels as spatiotemporal volumes within the XYT domain of a video. Here we show volumes for four face bounding boxes, for a time-aligned word from the video transcript, and for a commercial segment. Since the latter two labels have no inherent spatial bounds, they are represented by volumes spanning the full spatial extent of the video over the time range they occur.

They demonstrate how film scholars can use the interface to quickly find dialogue, scenes, and plot points in a single film or across the collection. MovieBrowser [Lehane et al. 2007; Mohamad Ali et al. 2011] further annotates films using computer vision techniques to extract shots and classify them as containing dialogue, action, or montages for subsequent search. VideoGrep [Lavigne 2018] retrieves source clips based on audio or subtitle search for use in supercuts. Like these systems, our queries also utilize aligned caption text and shot boundaries as spatiotemporal labels, but we provide a general way of programmatically creating new labels from these primitives.

Wu and Christie [2016] create a language for describing editing patterns in films, and use this language to find common editing idioms in a collection of hand-annotated film clips [Wu et al. 2017]. Both the lower-level video annotations used as built-in inputs to their system, as well as the higher-level film editing patterns themselves, can be more generally described as compositions of primitive video annotations in our system.

Content-based video query. The idea of procedurally defining video labels as compositions of existing labels goes back to the design of video query languages for early multimedia databases [Adali et al. 1996; Dönderler et al. 2004; Hibino and Rundensteiner 1995; Köprülü et al. 2002; Oomoto and Tanaka 1993]. These languages enabled video clip retrieval by specifying predicates in terms of desired temporal [Allen 1983] and spatial [Cohn et al. 1997] relationships between pre-annotated objects, but early systems were never used to analyze video collections at scale. Modern “data programming” [Ratner et al. 2017, 2016] is based on a mix of statistical and programmatic composition of annotations. However, we leverage composition to define high-level video labels, not increase the accuracy of existing noisy labels.

3 SPATIOTEMPORAL LABELS

We annotate the TV NEWS and FILM datasets with visual, audio, and text labels that serve as the basic primitives for analysis. Since these labels are multi-modal and sampled at different rates, we adopt a unified representation for all data items, which we call a *spatiotemporal label* (or just *label*). Such a label spans a continuous, axis-aligned volume of spacetime (X,Y,T) within a video. For example a detected face takes on its spatiotemporal bounds (e.g. “a face appeared from 2:15

to 2:18 within the box $\{x_1 : 0.2, x_2 : 0.1, y_1 : 0.4, y_2 : 0.6\}$). Labels with temporal, but no inherent spatial bounds, like a commercial segment, a word in a time-aligned transcript, or a single frame of video, span the full spatial (XY) domain. Fig. 2 illustrates multiple overlapping labels in a single video. We computed the following primitive labels on our datasets using Scanner [Poms et al. 2018]:

Color histograms. We compute 16-bin per-channel HSV color histograms on all video frames.

Face bounding boxes. We compute face bounding boxes using the MTCNN face detector [Zhang et al. 2016]. To reduce computation costs, we sample frames sparsely in time (every three seconds for TV NEWS, twice per second in FILM). The resulting labels have the XY extent of the detected box, and a time extent equal to the sampling period (e.g. 3 seconds for TV NEWS). In total we detect 344 million and 6.6 million faces in TV NEWS and FILM respectively.

Face embeddings. For each face we compute the 128-dimensional FaceNet descriptor [Schroff et al. 2015].

Face landmarks. For each face we compute face landmarks using the FAN model [Bulat and Tzimiropoulos 2017], which provides 68 2D keypoints of face features (jaw/nose/eyes/etc.).

Gender labels. For each TV News face we run a neural net gender classifier [Levi and Hassner 2015] to predict the person’s gender. We record the highest probability result (male, female) and the model’s confidence score. The neural net performed poorly on FILM due to makeup and costume, so for FILM we predict gender using a k-nearest neighbors classifier on face embeddings using a database of 14K hand-annotated gender-labeled faces from the dataset. Our final models are 91% and 96% accurate on TV NEWS and FILM respectively.

Person identities. For each show host (82 in total) as well as for select notable figures (e.g. political candidates) in the TV NEWS dataset, we train a binary logistic regressor to identify the individual from a face embedding. Each model is trained using thousands of hand-labeled exemplars from the TV NEWS dataset. For FILM we hand-label the face bounding boxes of leading characters for a selection of 90 of the 589 films (668 characters in total). See Section 4 of the supplemental material for a description of the labeling procedure in both cases.

Human pose. We computed human pose keypoints (14 2D locations of skeletal joints like neck/elbows/hips/etc.) using OpenPose [Cao et al. 2017]. Pose is computed on the same frames as face detection. The XY spatial extent of a human pose label is given by the bounds of its 2D keypoint locations. A total of 14.1 million poses were detected in FILM (pose was not run on TV NEWS due to concerns of cost efficiency).

Time-aligned transcripts. We used the Gentle aligner [Ochshorn 2018] to extract precise, sub-second word-level alignment from the raw transcripts. (Human-generated closed caption transcripts for TV NEWS provide a coarse time interval for 6-8 word phrases, not individual words, and provided timing information can sometimes be off by minutes). Each word in a transcript corresponds to a label that spans the exact period of time emitted by the aligner. 95% and 65% of the video transcripts were successfully aligned in TV NEWS

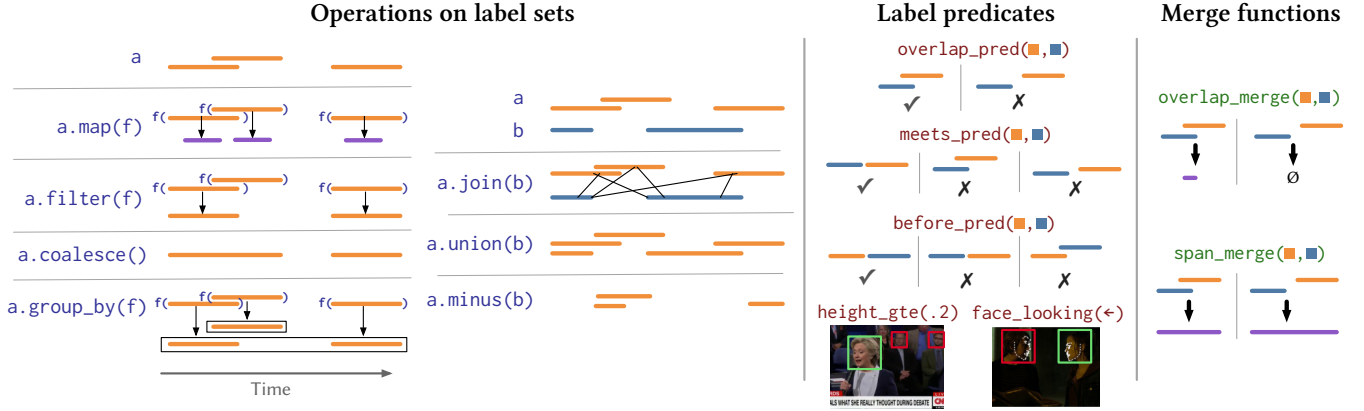


Fig. 3. Semantics of operations on labels and sets. The functions shown are a select sample of the complete library of spatiotemporal predicates and merging operations we use in our analysis, intended to provide visual intuition for how a few relevant operators work.

and FILM. (See Section 1 of the supplemental material for details full alignment procedure.).

Topic segments. We used a standard point-wise mutual information (PMI) method [Yang and Pedersen 1997] to generate a lexicon for a set of topic words chosen by the authors (immigration, presidential election, etc.), then classified segments based on the highest PMI-weighted sum of words in each lexicon (subject to a minimum threshold for segments with topics not in our selected set). Segments were generated from a sliding window of 60 seconds striding by 30 seconds, so the output is a series of equal length temporal labels annotated with the highest topic category.

Section 5 in the supplemental material contains further details about the accuracy of these primitive labels on our datasets, as well as additional primitive labels like hair color and topic segments that we computed but did not use in the in-paper analysis.

4 COMPOSING LABELS

The key idea of our approach is to construct higher-level labels via programmatic compositions of other labels. Consider the problem of labeling “women on screen during segments about immigration” in TV news. Our basic spatiotemporal labels (Section 3) include one for “women” (e.g. face gender is female) and one for the topic of “immigration”. Our desired higher-level label is the spacetime intersection of these sets. We can implement the intersection as a program using three kinds of compositional operators: functions to combine sets of labels, predicates to compare pairs of labels, and merge functions to combine label pairs together (Figure 3). We use a pseudocode language (Python-esque, but with types and concise inline functions) to present these concepts.

4.1 Spacetime Operations on Label Sets

To implement our example label “women on screen during immigration” as a program, the first step is to access label sets corresponding to female faces and to topic segments about immigration.

```
women : label set =
  database.table("faces").where("gender = female")
immigration_segments : label set =
```

```
database.table("topics").where("topic = immigration")
```

New labels are constructed by expressing relationships between labels within a single label set or between multiple label sets. For our example, we use the set operations `join`, `filter`, and `map`, to express transformations on the women and immigration segment label sets.

```
women_on_immigration : label set =
  women.join(immigration_segments)
    .filter(overlap_pred)
    .map(overlap_merge)
```

First, all pairs of women labels and immigration segments labels are created with a `join` (i.e. a cross product of the sets, as in a relational database). Then this set of label pairs is filtered using a predicate (`overlap_pred`) that tests whether a given woman label and immigration segment label overlap in spacetime. Finally, each woman/segment label pair that passes the predicate is transformed into a new label spanning their spacetime overlap using a merge function (`overlap_merge`), to produce an output label set that represents “women on screen during immigration segments”. Figure 4 illustrates how each of these steps in the algorithm manipulates intermediate label sets.

The `map` and `filter` operations are parameterized on functions used to manipulate or eliminate labels. In our example, a filter for `meets_pred` (whether a woman label is immediately followed by

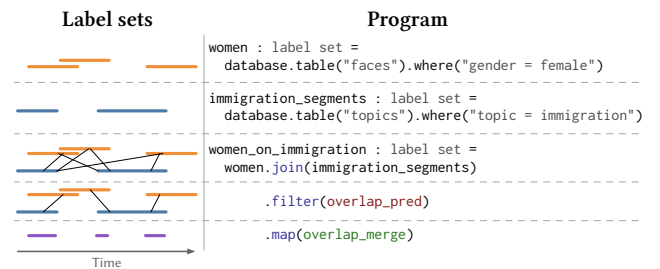


Fig. 4. Visualization of label sets at each point in the “women on screen during immigration segments” example. Orange bars are women labels, blue bars are immigration segment labels, and purple bars are the output labels.


```

1 # Commercial label algorithm
2 transcript_words = database.table("transcript_words")
3 histograms = database.table("histogram")
4
5 lower_case_words_sequences =
6     transcript_words
7     .filter(λ word → word.is_lowercase())
8     .coalesce(before_pred(max=5s) or after_pred(max=5s))
9
10 black_frame_sequences =
11     histograms
12     .filter(λ i → i.histogram.avg() < 0.01)
13     .coalesce(before_pred(max=0.1s) or after_pred(max=0.1s))
14     .filter(λ i → i.t2 - i.t1 > 0.5)
15
16 commercials =
17     lower_case_words_sequences
18     .join(black_frame_sequences)
19     .filter(after_pred(max=10s))
20     .map(span_merge)
21     .filter(λ i → 30 ≤ i.t2 - i.t1 ≤ 60*5)

```

Listing 1. To label commercials we identify lower case words in the transcript and sequences of black frames. Specifically we filter the transcript label set to retain only lower case words (lines 5-6). We then coalesce nearby lowercase words into continuous segments representing spans of time where the transcript is consistently lower case (line 7). Black frames are detected by averaging the bins of each channel in the HSV histogram (line 12). We then coalesce sequences of black frames denoting the start of commercial segments (line 13). The two sets are then joined, filtered and merged (lines 16-20) to form spans of lowercase word segments appearing within 10s after black frame sequences that form commercials.

an immigration segment label in time) and a map of `span_merge` (merging women/segment label pairs into a new label spanning both spacetime volumes) would return labels for “sequences of women followed by immigration segments.”

In practice, we provide a standard library containing a number of such functions (Figure 3). Set operations (blue) are mostly drawn from functions in stream processing systems and from standard set interfaces in functional programming languages. We include a variety of predicates like `overlap_pred` (highlighted red in code) for representing spacetime relationships between labels, like the temporal predicates *before*, *meets*, and *overlaps* (i.e. the Allen interval algebra [Allen 1983]) and 2D spatial predicates like *part of*, *tangent to* (i.e. the region connection calculus [Cohn et al. 1997]). For merge functions (green), we include `overlap_merge` for the spacetime intersection of two labels or `span_merge` for the spacetime union. Programmers can also write new predicates and merge functions as necessary for their tasks.

Non-local spatiotemporal relationships. Some labels are more naturally modeled through non-local reasoning between sets of spacetime volumes. For example, non-local spatiotemporal relationships play a key role in labeling commercials in TV NEWS. We have found that for our dataset, a combination of visual and text primitive labels yields an accurate commercial labeler. Specifically, we identify commercials heuristically as *segments between 30 seconds and 5 minutes that contain lower case words in the transcript and start with a sequence of at least 0.5 seconds of black video frames*. The corresponding code is shown in Listing 1.

```

1 # Reaction shot label algorithm
2 named_faces = database.table("face").join("names")
3 script_lines = database.table("script_lines")
4 closeup_shots = database.table("shot").where("type = closeup")
5
6 speaker_on_screen =
7     script_lines
8     .join(named_faces).filter(overlap_pred).map(span_merge)
9     .group_by(λ line → line.t1)
10    .filter(λ (t1, line_faces) →
11        line_faces.filter(λ (line, face) → line.name == face.name)
12            .length() ≥ 1)
13    .map(λ (_, line_faces) → line_faces[0])
14
15 shots_with_speaker_on_screen =
16     closeup_shots
17     .join(speaker_on_screen).filter(overlap_pred).map(span_merge)
18
19 shots_with_speaker_off_screen =
20     closeup_shots
21     .join(script_lines).filter(overlap_pred).map(span_merge)
22     .minus(shots_with_speaker_on_screen)

```

Listing 2. Computing reaction shots with grouping. First, all the faces that were on screen during delivery of a given script line are grouped together (lines 6-8). The lines are filtered to keep ones where the character speaking has the same name as one of the on-screen faces (lines 9-12). Then the script lines are matched with their containing shot labels (lines 14-16). Finally all shots that lack dialogue (lines 19-20) and those that have a speaker on screen (line 21) are removed, leaving the reaction shots.

Listing 1 introduces a `coalesce` operator that iteratively merges all pairs of labels in a set that pass a given predicate (Figure 3 shows a visual example). This operation requires non-local reasoning in the sense that an arbitrary number of labels can be coalesced at once, as opposed to merging only pairs of labels in a join.

To exclude commercials from analysis, e.g. in labeling “women on screen during immigration segments and not during commercials”, we need to remove all parts of the composite label that overlap with commercials. If we interpret a label set not as a set of discrete label objects, but instead a set of continuous points in spacetime, then this operation logically corresponds to a set difference operation, which we call `minus` (also shown in Figure 3).

```

women_on_immigration_no_commercials =
    women_on_immigration.minus(commercials)

```

Nested operations through groups. Many of our analyses contain notions of labels that are grouped together along an axis in time or space. For example, in FILM, a “reaction shot” is a *close-up shot where the current speaker is not on screen*. As shown in Listing 2, to produce labels for reaction shots, we use a `group_by` operator. `group_by(λ line → line.t1)` groups together all labels with the same start time. This operation groups all the faces on screen during the delivery of a given line. More generally, `group_by` groups all set elements with the same key given a function that generates a key for each element. The combination of `group_by` and `map/filter` permit arbitrary aggregations over groups, which is more expressive than fixed function aggregations like `sum`, `count`, and `average` used in conjunction with a SQL GROUP BY.

```

1 # "Hermione positioned between Harry and Ron" label algorithm
2 hermione_pattern = [
3   (["hermione"], λ fs → fs[0].name == "Hermione"),
4   (["ron"], λ fs → fs[0].name == "Ron"),
5   (["harry"], λ fs → fs[0].name == "Harry"),
6   (["hermione", "ron", "harry"], λ fs →
7     fs[1].x1 ≤ fs[0].x1 ≤ fs[2].x1 or fs[2].x1 ≤ fs[0].x1 ≤ fs[1].x1)
8   (["hermione", "ron", "harry"], λ fs →
9     ∀ i ∈ 0..2, |fs[i].y1 - fs[i+1].y1| < 0.1)
10 ]
11
12 named_faces = database.table("face").join("names")
13 hermione_labels =
14   named_faces
15   .group_by(λ face → face.t1)
16   .filter(λ (_, frame_faces) →
17     frame_faces.matches(hermione_pattern).length() > 0)

```

Listing 3. Finding the Harry Potter trio with spatial patterns. The first three predicates identify faces corresponding to our characters (lines 3-5). The next predicate checks that Hermione (fs[0]) is between Harry and Ron (lines 6-7). The last checks that all three are close in the y axis (lines 8-9). We run the pattern over all groups of faces shown at the same time (i.e. in the same frame).

4.2 Patterns in Time and Space

While set operations are sufficient to implement the labels used in our analysis, describing labels involving many spatiotemporal relationships can be tedious. Several of our labels involve looking for subsets of intervals that collectively satisfy a complex set of predicates, i.e. that match a pattern. Patterns may involve temporal relationships such as shot scale sequences [Wu et al. 2018], e.g. a “zoom out” in FILM is a *medium shot followed by a long shot*. In practice, we find it useful to implement these patterns using a pattern matching language built on top of our compositional operators. We use a relational search interface inspired by logic programming languages like Datalog [Ceri et al. 1990].

```

zoom_out_pattern = [
  (["shot1"], λ shots → shots[0].scale == "medium"),
  (["shot2"], λ shots → shots[0].scale == "long"),
  (["shot1", "shot2"], λ shots → meets_pred(shots[0], shots[1]))
]

Shots = database.table("shots")
zoom_out_sequences : interval list set =
  shots.matches(zoom_out_pattern)

```

Given a label set of shots annotated with shot scales (see Section 5.2 for how to compute shots and shot scales), we write the “zoom out” pattern with three predicates: two shot scale tests check that the input shot is either medium or long, and a `meets_pred` checks if the two shots are adjacent in time. The `matches` function then returns a set of all pairs of shots that satisfy those constraints.

Generally, a pattern is defined by a set of variables that correspond to labels in a set, and a set of predicates describing required relationships between these labels. Matching a pattern amounts to finding a subset of labels for which all required relationships hold. Listing 3 illustrates a more complex example involving spatial patterns for detecting *Hermione positioned between Harry and Ron* (as visualized in Figure 1).

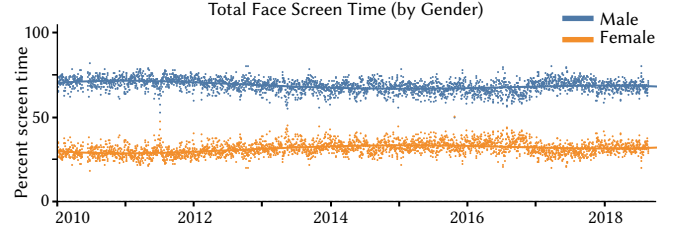


Fig. 5. Women receive 31.2% of the face screen time on cable TV news. Each dot represents a daily percentage.

5 VIDEO ANALYSIS RESULTS

We have used compositions of spatiotemporal labels and patterns to analyze the TV NEWS and FILM datasets. We highlight a few key findings here and present additional analyses in the Section 2 of the supplemental material.

5.1 Analysis of Cable TV News

31% of the overall total face screen time is women. We computed screen time for male and female faces across the entire TV News dataset (Fig. 5). Overall, women receive 31% of the total face screen time. This proportion increases from 29% in 2010 to a peak of 34% in 2016 before dropping to 32% in 2018. This trend is similar across all three news networks.

We estimate gender screen time by totaling the time span of all face labels for a particular gender. For example, two female face bounding boxes appearing on screen for 6 seconds contributes 12 seconds to total female screen time. We exclude all face labels that intersect with commercial labels (Section 4) from the screen time estimate, since our goal is to focus on the presentation of news content. Additionally, we filter faces with a bounding box height less than 20% of the frame size to ignore small faces that are likely not the focus of news segments (e.g. crowds at a rally, pedestrians in B-roll footage). All subsequent analyses in this section exclude commercials and small faces.

Female hosts constitute an increasing fraction of host screen time. We differentiate screen time of hosts from non-hosts using face identity labels and plot host screen time by gender in Fig. 1-left. The fraction of female host screen time increased from 30% in 2010 to 38% in 2018, peaking at 42% in 2015. Fig. 6 provides a per-channel view of the data, and clearly illustrates the departure of several key female hosts from Fox News in late 2016 (Megyn Kelly, Gretchen Carlson, Greta Van Susteren).

Select hosts dominate screen time of their shows. Given the popularity of leading cable TV news hosts, we can create a “host narcissism index” by computing the fraction of time a host is on screen on their own show. Fig. 7 plots this quantity. Hosts appear on screen for an average 24% ($\sigma=12\%$) of their show, but a few outliers emerge: Megyn Kelly and Brooke Baldwin are on screen an average 46%, Tucker Carlson for 62%, and Chris Cuomo has by far the most average per-show screen time at 73%.

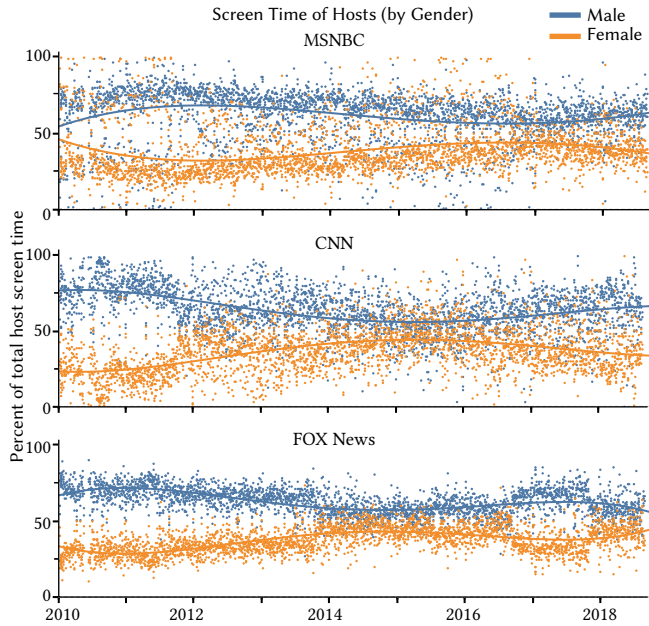


Fig. 6. The percentage of female host screen time increased from 30% to 38% since 2010. The sudden decrease in late 2016 Fox female host screen time is due to the departure of a number of female hosts.

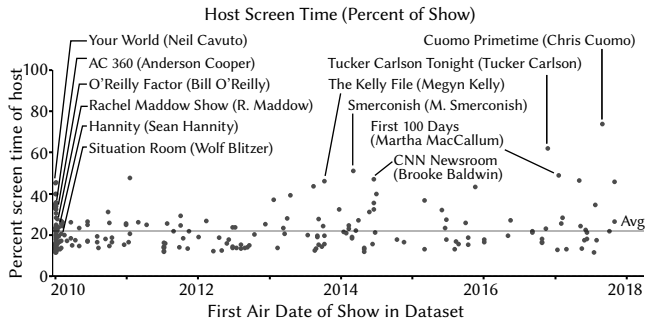


Fig. 7. A few hosts have substantially higher screen time on their own shows compared to other hosts. Each dot is the average percentage of total video time that a host is on screen during their show.

Interviewees get the majority of screen time on interviews. The interview is an important moment on TV news, so we developed temporal patterns (Section 4.2) that detect interview segments. These patterns match sequences of either the host, the interviewee, or both individuals appearing on screen. We used these patterns to identify a number of interviews with 40 prominent politicians: presidential candidates, ranking members of Congress, and others (see Section 3 in the supplemental material for details). On average, the interviewees appear on screen alone for 51% of the time during these interviews (an additional 15% of the time they appear along with the host). By contrast, hosts appear alone on screen for only 12% of the interview, suggesting that on average they cede most

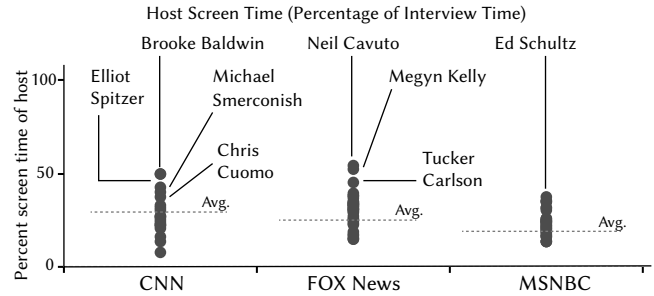


Fig. 8. A few hosts also have higher screen time in interviews. Each dot is the average percentage of total interview time that a host is on screen.

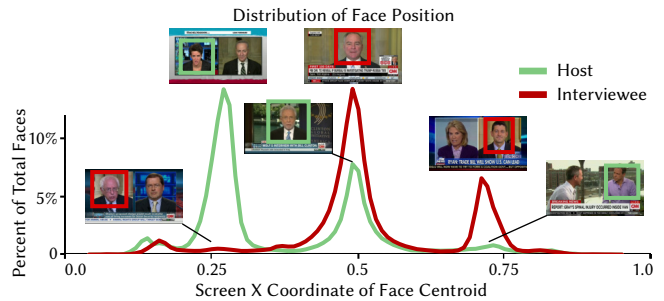


Fig. 9. Hosts typically appear on the left side of the screen during interviews, with guests displayed at center or on the right.

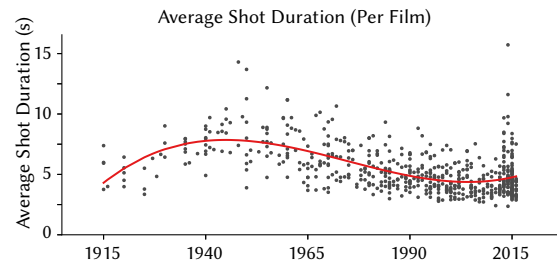


Fig. 10. The average shot duration in films has decreased over time. We show linear and cubic fits to mimic prior analyses by [Cutting et al. 2011]. Each data point is a film.

screen time to the guest. These ratios are similar across channels as well as across interviewees.

Some hosts do not leave the screen even in interviews. Fig. 8 extends our narcissism index by plotting host screen time during their interviews of the selected politicians. Although on average hosts are on screen for only 12% of their interviews, some hosts rarely go out of view. Consistent with the trends in Fig. 7, Brooke Baldwin, Tucker Carlson, and Megyn Kelly all are on screen (shown by themselves or with the interviewee) for over 45% of the duration of their interviews.

Hosts are on the left, guests are on the right. While common practice for TV talk shows is to keep the host on the right side of the

screen during interviews [Malone 2010], cable TV news takes the opposite approach. Fig. 9 shows the distribution of face positions on the X-axis of the video frame for the hosts and guests during interviews. Across every network and host, the spatial framing of interviews appears to follow a consistent trend: hosts are on the left or middle of the frame 94% of the time, while guests are on the middle or right of the frame 91% of the time.

5.2 Analysis of a Century of Film

Prior studies of film editing style focus on the properties of shots (a segment of continuous footage between cuts in an edited film): what is their duration, how are shots framed (close up or wide angle), and how have these characteristics evolved over time?

Shots have been getting shorter in duration. Using manual annotation of a small number of films, film scholars have noted the shrinking duration of shots as the pace of films has increased [Cutting et al. 2011; Cutting and Candan 2015]. As shown in Fig. 10, we observe the same trend when performing an automated analysis of all 589 films. Average shot duration creeps upward in the final years of the FILM dataset due to several films with very long shot durations, such as *Birdman* (average length 15.7 seconds), *12 Years a Slave* (9.7 seconds), and *The Revenant* (8.4 seconds).

To carry out this analysis we construct a robust shot detection algorithm by composing HSV histogram and face bounding box labels. The algorithm first uses frame-to-frame differences in HSV histograms to partition a video into segments that exhibit little visual change. Some of these initial segment boundaries correspond to moments of significant scene or camera motion, not cuts in the edited film. To reduce these false positive boundaries we compute face bounding boxes in the frames before and after each segment boundary and coalesce segments where the number and position of faces on both sides of their boundary match. See Sections 3 and 5 of the supplemental material for a full algorithm description and validation of accuracy against hand-labeled shot boundaries.

Prior work also identified trends in shot length across the acts of a film [Cutting 2016] (e.g. shots become shorter during tense moment's of a film's climax). We observe similar trends in the FILM dataset. (See Section 2 of the supplemental material.

Shots have been getting closer in scale. Shot scale is a measure of the field-of-view of a shot, as is often described in terms of seven categories: long, long, medium long, medium, medium close up, close up, and extreme close up [Cutting and Armstrong 2016]. On hand annotated datasets, film scholars have observed increasing use of closer shot scales over time [Cutting and Armstrong 2016]. As shown in Fig. 11, consistent with prior observations, use of medium close ups, close ups, and extreme close ups has increased over the duration of the FILM dataset, whereas the medium, medium long, long, and extreme long shots have become less common.

To compute shot scale we find all people within the temporal range of a shot (via face bounding boxes or the orientation and size of a pose) and estimate scale from the largest of these detections. We adopt same shot scale categorization as [Cutting and Armstrong 2016]. Sections 3 and 5 of supplemental materials includes a full shot scale algorithm definition and a qualitative evaluation of accuracy.

In supplemental we also present additional shot-based studies that have appeared in prior research on film style, including: studies of the number of actors on screen over time and the correlation of shot length to the number of actors on screen. [Cutting 2015].

Male actors receive nearly three times as much screen time as female actors. Across our entire dataset (excluding animated films), average female screen time was only 26.3%. As shown in Fig. 12 female screen time decreased until 1975 and has been slowly increasing since. The film with the highest percentage of female screen time was *Gravity* (80.3% female). *Locke*, which takes place entirely inside a car with a single male actor, had the least female screen time. 115 films in our dataset featured a female lead as the first actor in the credits list; in these films average female screen time was 46.9%, almost achieving gender parity. Notably, there have been more films featuring a large fraction of female screen time in recent years.

Most of the male/female screen time disparity is due to films containing a greater number of male faces. However, we also found that shots containing all male faces were on average 16.8% longer in duration than exclusively female faces. See supplemental material for more details on this analysis and other analysis of female screen time, including analysis of female screen time by genre.

Conversations between characters constitute just over 40% of film time. Conversations between a film's characters drive the narrative of a film, so we create patterns that label conversations in our films. These patterns, which utilize shot labels and face embeddings, begin by finding consecutive shots involving two different characters. We then incrementally construct labels for full conversation segments by iteratively coalescing pairs of shots that overlap in time and contain the same two characters. (See supplemental for details on how this algorithm also works to find conversations among N-persons.) Using this algorithm we find that conversations constitute 41.7% of our films.

Labeling conversational segments provides the opportunity to create further patterns that identify common film editing idioms for dialog-driven scenes. For example, as shown in Fig. 13 and 14 we have developed patterns for *intensification* (i.e. shot scale moves in closer on an actor over the course of a conversation) and *start wide* (i.e. an establishing shot that occurs before a conversation). We find that 7.9% of conversations follow the intensification idiom, and 9.4% follow the start wide idiom.

6 VIDEO SYNTHESIS RESULTS

Composing spatiotemporal labels enables us to search for a wide variety of patterns across a video collection. We find that visualizing labels as supercuts (videos that play each clip in sequence) and/or montages (a grid of clips that play together on screen) can help us make sense of the different forms a spatiotemporal pattern may take. All the examples in this section are available in the web page contained in supplemental material.

For example, a supercut for the spatial pattern of Hermione positioned between Harry and Ron (Listing 3, Fig. 1) shows that this is a recurring visual template across the Harry Potter series of films and appears with different background, lighting, and emotional points. Composing this pattern with shot scale labels shows that the trio

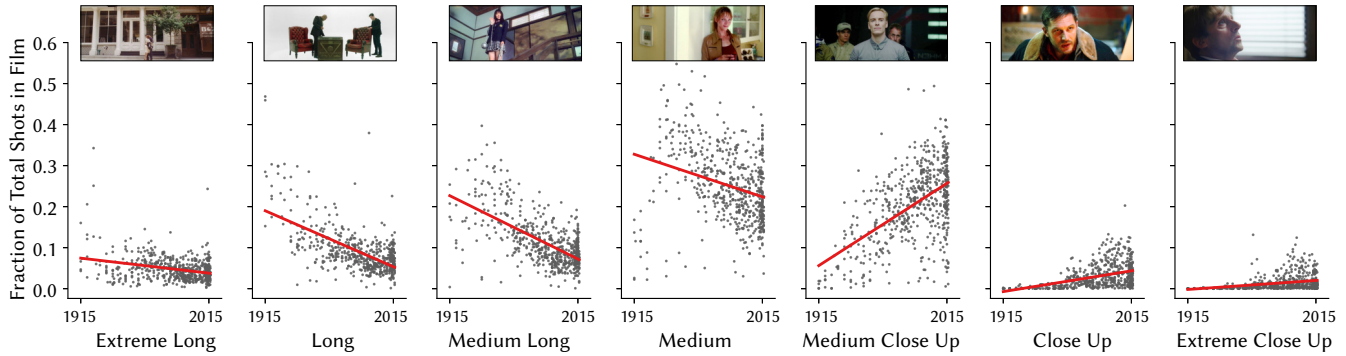


Fig. 11. The scale of shots has moved “closer” over time. Use of medium and long shots has been decreasing, while usage of tightly framed shots (medium close up, close ups, and extreme close ups) has increased.

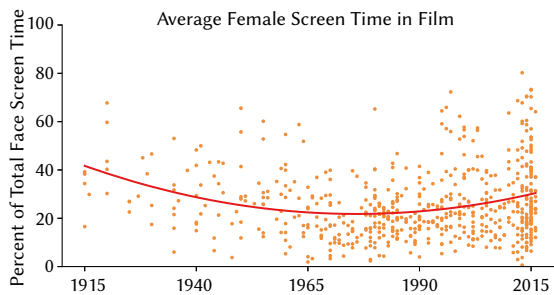


Fig. 12. Male actors receive nearly three times as much screen time as female actors in our FILM dataset, although the fraction of female screen time has on average increased since 1975.

were increasingly shown with longer shot scales in the later films, until the final resolution at the end of the final film in the series.

We have similarly generated a supercut of action sequences, which we define as sequences of five or more short shots (each shot is less than 0.8s in duration). The resulting clips show a variety of quick cuts from car chases to fight scenes and monster attacks. A supercut of reaction shots (Listing 2) in the film *Apollo 13* finds shots where the person identity for the face onscreen is different from the name of the person delivering the line in the script. Such reaction sequences show astronauts receiving orders from ground control, or characters reacting to television broadcast, for example. These shots can emphasize the emotional impact of certain lines by showing a main character’s reaction.

Montages can simultaneously reveal spatiotemporal patterns across multiple clips. For example we created a montage of interviews between hosts and presidential candidates in 2012 and 2016 which shows that hosts almost always appears on the left when both the host and guest are onscreen (Fig. 1-center). Similarly we created a montage of people on TV News saying the phrases “Happy New Year” and “Thank you very much”.

We can use these query results as a starting point and manually curate the resulting clips to synthesize cover songs by people in our videos. For example, we can synthesize Barack Obama singing

“Happy Birthday.” Given the lyrics of the song along with closed caption timings for TV News transcripts, we compose the transcript and person identity labels to find each time Barack Obama says each of the word in the song. We automatically rank each resulting clip by its difference in length from the corresponding word’s closed caption timings in the song. Selecting the top ranked candidate automatically generates a cover of the song “sung” by Obama. In practice we have found that a small amount manual clip curation among the top ranked candidates further improves results and allows us to quickly produce results similar those found on the BaracksDubs YouTube channel [Saleh 2018]. Figure 15 shows an example of Barack Obama singing a line from Johnny Cash’s *Folsom Prison Blues*.

7 DISCUSSION

Comparison with machine learning methods. The main idea in this paper is to construct high-level labels through procedural composition. This approach contrasts with current trends in modern machine learning, which pursue end-to-end learning from raw data (e.g. pixels, text) based on hand-labeled examples [Hassanien et al. 2017; He et al. 2017; Mai et al. 2017]. When it works, building labels through procedural composition lets users go from a hypothesis to a set of labeled results in minutes. Such a program does not need additional training data (only pretrained basic primitives), allows a user to express heuristic domain knowledge (via programming), modularly build on other labels, and debug failure modes.

However, procedural composition also has limits—higher-level composition is difficult when lower-level labels do not exist or fail in a particular context. For example, we have been unable to build a reliable detector for kissing scenes because face and pose detectors fail due to occlusions during an embrace. Also, more complex labels (e.g. interviews in TV News) involve manual parameter tuning to correctly set overlap or distance thresholds for our datasets. However, we conjecture that the interpretability of the parameters involved in modular composition make them easier to adjust than model architecture or learning rate hyperparameters in end-to-end learning systems. We leave it to future work to investigate this hypothesis. Nevertheless, going forward, we believe a system for exploring visual data should allow the user to move incrementally along a spectrum from programs to statistical models by refining



Fig. 13. Two examples of detected conversations that exhibit the "intensification" idiom, from *Spellbound* and *Braveheart*. Shot scale moves in closer on one or both actors over the course of a conversation. In *Spellbound*, shot scale moves closer on the male lead in the third and fifth shots and on the female lead in the sixth shot. In *Braveheart*, shot scale moves closer on the female lead in the third shot and on the male lead in the fourth shot.



Fig. 14. Two examples of detected conversations that exhibit the "start wide" idiom, from *Star Wars: Episode VI - Return of the Jedi* and *Interstellar*. Shot scale starts with a long shot before cutting in closer to a conversation.

labels with both heuristics and training data [Johnson et al. 2017; Ratner et al. 2016].

Ethical concerns. As systems like ours make it increasingly possible to mine rich personal data from video, we must consider carefully the implications of the data we compute and the ways we use it. For example, should we attempt to predict attributes like a person's gender from their picture in the first place? Gender is a complex biological and psychological phenomenon, which a classifier reduces to (effectively) visual stereotypes—women have long hair, men are bald, women wear lipstick, and so on. We use visual gender classifiers in these analyses because we believe that understanding gender balance in TV NEWS and FILM is important and that the classifiers are necessary in order to do our analysis on the scale needed. Our predicted gender will certainly not always match everyone's true

gender preference, but in aggregate it is correct enough to produce meaningful insights.

New opportunities for visual data analysis and synthesis. In the same way N-gram analysis on centuries of books promotes an understanding of culture through large-scale data analysis [Michel et al. 2011], we believe that analysis of large video datasets like TV NEWS and FILM can help quantify the content and presentation of modern video media. Intelligent video synthesis presents opportunities to manage and summarize the enormous volume of information contained in modern visual media. We are excited to see further work in developing infrastructure and query languages to make visual analysis at this scale more accessible to everyone.

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Fig. 15. A line from our supercut of Obama singing Johnny Cash's *Folsom Prison Blues*.

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